MET 302 Dynamics Midterm exam

Name: _____

1. A mountain biker starts from rest at a trailhead. As she descends, her GPS records her position in meters as a function of time as

$$x = 0.75t^3 + 0.5t^2$$

Determine her position, velocity, and acceleration when $t = 3$ seconds.

Question 1 Answers:

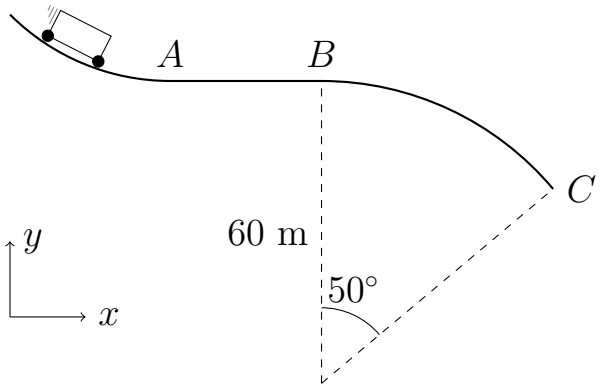
$x =$ _____ m $v =$ _____ m/s $a =$ _____ m/s^2

2. A shot put is released from a height of 1.6 m with an initial speed of 14 m/s at a 40° angle above the horizontal. How far, in meters, does the shot travel? What is the shot's overall speed when it contacts the ground? Use $g = 9.81$ m/s^2.

Question 2 answers:

$x =$ _____ m $v =$ _____ m/s

3. A roller-coaster car travels at a speed of 26 m/s when it passes through point B. At that point, it enters a concave down circular section of the track that has a radius of curvature of 60 m. As it travels along the curve from B to C, the car's speed increases at a constant rate of 3 m/s^2. Determine the time it takes the car to reach point C, and the x and y components of the overall acceleration of the car when it reaches point C.



Question 3 Answers:

$t =$ _____ s $\quad a_x =$ _____ m/s^2 $\quad a_y =$ _____ m/s^2